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MILITARY STANDARD

LEADS FOR ELECTRONIC COMPONENT PARTS

AMSC N/A

FSC 5999

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DEPARTMENT OF DEFENSE
Washington, DC 20360

Leads for Electronic Component Parts

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Laboratory Command, Industrial Engineering and Development Division, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5302 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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CONTENTS

Paragraph			Page
	1.	SCOPE	1
	1.1	Purpose	1
	1.2	Classification	1
	1.2.1	Type designation	1
	1.2.1.1	Cross section	1
	1.2.1.2	Nominal size	1
	1.2.1.3	Composition	1
	1.2.1.4	Final finish	1
	1.2.1.5	Underplating	2
	2.	REFERENCED DOCUMENTS	4
	2.1	Government documents	4
	2.1.1	Specifications and standards	4
	2.2	Other publications	4
	2.3	Order of precedence	5
	3.	DEFINITIONS (Not applicable)	6
	4.	GENERAL REQUIREMENTS	7
	4.1	Preferred methods of assembly	7
	4.2	Solderability	8
	4.3	Thickness	8
	4.4	Adhesion	8
	4.5	Porosity	8
	4.6	Workmanship	8
	5.	DETAILED REQUIREMENTS	9
	5.1	Composition	9
	5.2	Final finish	9
	5.2.1	Underplating requirements	10
	5.2.1.1	Gold-plating	10
	5.2.1.2	Compositions F, G, K	10
	5.2.1.3	Hot solder dip	10
	5.2.1.4	Compositions A, B, J, X	10
	6.	NOTES	11
	6.1	Precaution	11
	6.2	Finishes	11
	6.3	Removal of gold-plating	11
	6.4	Passive nickel	11
	6.5	Cross-reference	11
	6.6	Tin-melting	12
	6.7	Cleanliness	12
	6.8	Subject term (key word listing)	12
	6.9	Changes from previous issue	12
		TABLES	
	Table I	Nominal size	3
	Table II	Preferred methods of assembly	7
	Table III	Cross-reference of lead types	11

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1. SCOPE

1.1 Purpose. This standard establishes the method of identifying, specifying, and limiting to a minimum the combinations of basic materials, the thickness and composition of the plating used therewith, and the sizes of wire and flat wire (ribbon) lead on electronic and electrical parts when they are to be soldered or welded during assembly in equipment.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:

L	W	(X.XX)	D	-	21	-	A1
Lead	Cross-section (see 1.2.1.1)	Nominal size (see 1.2.1.2)	Composition (see 1.2.1.3)	Dash	Final finish (see 1.2.1.4)	Dash	Under-plating (see 1.2.1.5)

1.2.1.1 Cross-section. The lead cross-section shall be identified by a single letter symbol as follows:

R - Rectangular cross-section.
S - Square cross-section.
W - Wire (round or circular cross-section).
X - Other cross-sections as specified.

1.2.1.2 Nominal size. The lead size shall be identified by the nominal diameter or nominal diagonal in millimeters within parentheses or inches without parentheses. See table I for a list of suggested nominal sizes.

1.2.1.3 Composition (see 5.1). The lead composition shall be identified by a single letter symbol as follows:

A - Copper-iron-zinc alloy (194 alloy).
B - Copper-iron-tin alloy (195 alloy).
C - Copper (oxygen free).
D - Iron-nickel alloy, copper-clad (Dumet).
E - Copper-nickel-tin alloy (725 alloy).
F - Iron-nickel alloy (52 alloy).
G - Iron-nickel alloy (42 alloy).
H - Copper-aluminum-silicon-cobolt alloy (638 alloy).
J - Copper-zinc-aluminum-cobolt alloy (688 alloy).
K - Iron-nickel-cobolt alloy (per ASTM F-15).
L - Copper (electrolytic tough pitch).
M - Steel, nickel-clad, 8 percent by weight nickel cladding.
N - Nickel.
W - Steel, copper-clad.
X - Other compositions as specified in the applicable acquisition document.

1.2.1.4 Final finish (see 5.2). The specific final finish for leads shall be identified in the following manner:

01 - No finish.
21 - Gold plating, electrodeposited, 100 microinches minimum to 300 microinches maximum.
22 Gold plating, electrodeposited, 50 microinches minimum to 100 microinches maximum.
31 - Acid-tin plating, electrodeposited, 300-500 microinches.
32 - Tin-lead plating, electrodeposited (as deposited).
33 - Matte-tin plating (low organic content), electrodeposited, 300-500 microinches.
34 - Fused (reflowed) tin-lead plating, electrodeposited.
35 - Acid-tin plating, electrodeposited, 100-300 microinches.
36 - Matte-tin plating (low organic content), electrodeposited, 100-300 microinches.

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51 - Hot solder dip, 200 microinches minimum (measured at the crest).
52 - Hot solder dip, 60 microinches minimum, 200 microinches maximum (measured at crest).
61 - Nickel plating, electrodeposited.
71 - Silver plating, electrodeposited, type II, grade B (per QQ-S-365).
72 - Silver plating, electrodeposited, type I, matte, nonbrightened (per QQ-S-365).

1.2.1.5 Underplating (see 5.2.1). The type of underplating shall be identified in the following manner:

A - Nickel, electroytic.
  1 - Thickness: 60 microinches (1.52 µm) minimum to 150 microinches (3.81 µm) maximum.
  2 - Thickness: 100 microinches (2.54 µm) minimum to 300 microinches (7.62 µm) maximum.
  3 - Thickness: 20 microinches (0.50 µm) minimum to 60 microinches (1.52 µm) maximum.
B - Nickel, electroless (not to be used on flexible or semirigid leads).
  1 - Thickness: 100 microinches (2.54 µm) minimum to 200 microinches (5.08) maximum.
  2 - Thickness: 50 microinches (1.27 µm) minimum to 120 microinches (3.04 µm) maximum.
C - Copper, electrodeposited.
  1 - Thickness: 60 microinches (1.52 µm) minimum to 150 microinches (3.81 µm) maximum.
  2 - Thickness: 150 microinches (3.81 µm) minimum to 300 microinches (7.62 µm) maximum.
  3 - Thickness: 20 microinches (0.50 µm) minimum to 60 microinches (1.52 µm) maximum.
P - Palladium-nickel alloy.
  1 - Thickness: 50 microinches (1.27 µm) minimum to 100 microinches (2.54) maximum.
N - No underplating.

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TABLE I. Nominal size (for information only).

Wire size AWG (B and S)	Diameter in inches	Diameter in millimeters
1	0.289	7.34
2	0.257	6.53
3	0.229	5.82
4	0.204	5.18
5	0.181	4.60
6	0.162	4.11
7	0.144	3.65
8	0.128	3.25
9	0.114	2.90
10	0.101	2.57
11	0.090	2.29
12	0.080	2.03
13	0.072	1.83
14	0.064	1.63
15	0.057	1.45
16	0.050	1.27
17	0.045	1.14
18	0.040	1.02
19	0.035	0.89
20	0.032	0.81
21	0.028	0.71
22	0.025	0.64
23	0.022	0.56
24	0.020	0.51
25	0.017	0.43
26	0.015	0.38
27	0.014	0.36
28	0.012	0.30
29	0.011	0.28
30	0.010	0.25
31	0.008	0.20
32	0.008	0.20
33	0.007	0.18
34	0.006	0.15
35	0.005	0.13
36	0.005	0.13
37	0.004	0.10
38	0.004	0.10
39	0.003	0.08
40	0.003	0.08

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2. REFERENCED DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-S-365 - Silver Plating, Electrodeposited; General Requirements for.
QQ-S-571 - Solder, Tin Alloy: Tin-Lead Alloy; and Lead Alloy.

MILITARY

MIL-T-10727 - Tin Plating, Electrodeposited or Hot-Dipped, for Ferrous and Nonferrous Metals.
MIL-C-14550 - Copper, Electrodeposited.
MIL-C-26074 - Coatings, Electroless Nickel, Requirements for.
MIL-G-45204 - Gold Plating, Electrodeposited.
MIL-N-46025 - Nickel Bar, Flat Wire (Ribbon) and Strip (for Electronic Use).
MIL-N-46026 - Nickel, Rod and Wire (Round) (for Electronic Use).
MIL-P-81728 - Plating, Tin-lead (Electrodeposited).

STANDARDS

MILITARY

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the issue of the DODISS specified in the solicitation. The issues of documents which have not been adopted shall be those in effect on the date of the cited DODISS.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B5 - Electrolytic Tough-Pitch Copper Refinery Shapes, Specification for.
ASTM B122 - Copper-nickel-tin alloy, Copper-nickel-zinc alloy (Nickel silver), and Copper-nickel alloy, Plate, Sheet, Strip and Rolled Bar.
ASTM B170 - Oxygen-Free Electrolytic, Copper-Refinery Shapes.
ASTM B322 - Cleaning Metals Prior to Electroplating.
ASMT B343 - Preparation of Nickel for Electroplating with Nickel.
ASTM B465 - Copper-Iron Alloy Plate, Sheet, Strip and Rolled Bar, Specification for.
ASTM B487 - Measurement of Metal and Oxide Coating Thickness by Microscopical Examination of a Cross-section, Method for.
ASTM B571 - Adhesion of Metallic Coating, Test Methods for.
ASTM B583 - Porosity in Gold Coatings on Metal Substrates, Test Methods for.
ASTM B592 - Copper-Zinc-Aluminum-Cobalt or Nickel-Alloy Plate, Sheet, Strip and Rolled Bar, Specification for.
ASTM F15 - Iron-Nickel-Cobalt Sealing Alloy, Specification for.
ASTM F29 - Dumet Wire for Glass-to-Metal Seal Applications, Specification for.
ASTM F30 - Iron-Nickel Sealing Alloy, Specification for.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this standard and the reference cited herein, the text of this standard shall take precedence.

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3. DEFINITIONS

3.1 Definitions. Not applicable.

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4. GENERAL REQUIREMENTS

4.1 Preferred methods of assembly. The preferred methods of assembly shall be in accordance with those listed in table II.

TABLE II. Preferred methods of assembly.

Composition	Underplating	Final finish											
		01	21	22	31	32	33	34	51	52	61	71	72
A	A	---	S,W	S,W	S	S	S	S	S	S	W	S	S
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
B	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
C	A	---	---	---	"	"	"	"	"	"	"	"	"
"	B	---	---	---	"	"	"	"	"	"	"	"	"
"		---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
"	N	W	---	---	"	"	"	"	"	"	"	"	"
D	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
"	N	---	---	---	"	"	"	"	"	"	"	"	"
E	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
"	N	---	---	---	"	"	"	"	"	"	"	"	"
F	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
G	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
H	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
"	N	---	---	---	"	"	"	"	"	"	"	"	"
J	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
K	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
L	A	---	---	---	"	"	"	"	"	"	"	"	"
"	B	---	---	---	"	"	"	"	"	"	"	"	"
"					"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
"	N	---	---	---	"	"	"	"	"	"	"	"	"
M					"	"	"	"	"	"	"	"	"
"					"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
"	N	---	---	---	"	"	"	"	"	"	"	"	"

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TABLE II. Preferred methods of assembly--- Continued.

Composition	Underplating	Final finish											
		01	21	22	31	32	33	34	51	52	61	71	72
N					S	S	S	S	S	S	W	S	S
"					"	"	"	"	"	"	"	"	"
"	C	---	---	---	"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
"	N	W	---	---	"	"	"	"	"	"	"	"	"
W	A	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"	B	---	S,W	S,W	"	"	"	"	"	"	"	"	"
"					"	"	"	"	"	"	"	"	"
"	P	---	---	---	"	"	"	"	"	"	"	"	"
"	N	---	---	---	"	"	"	"	"	"	"	"	"
X	As specified in acquisition document												

S = Solder

W = Weld

4.2 Solderability. When specified on the acquisition document, leads shall meet the solderability requirements as follows:

a. Discrete semiconductor devices: Test method 2026 of MIL-STD-750 shall be used.

b. Microelectronics: Test method 2002 or test method 2003 of MIL-STD-883 shall be used.

c. Other electronic and electrical component parts: Test method 208 of MIL-STD-202 shall be used.

4.3 Thickness. When specified on the acquisition document, the thickness of the final finish and underplating shall be determined in accordance with ASTM B487 or ASTM B567.

4.4 Adhesion. When specified on the acquisition document, the adhesion of the final finish and underplating shall be determined in accordance with ASTM B571 or method 2025 of MIL-STD-883.

4.5 Porosity. When specified on the acquisition document, there shall be no evidence of porosity for final finishes 21, 22, and 61 when tested in accordance with ASTM B583.

4.6 Workmanship in finishes. All plating of primary finishes or underplating shall be deposited in such a way that the plating is applied to a clean, nonoxidized metal surface that is free from defects that will be detrimental to the utility, form, fit, function, or protective value of the finish. Where surface cleaning, deoxidation, activation, undercoating, etc. are used, the material or device so treated must proceed immediately to the primary finish process, in order to maintain the ability to overplate effectively (see 6.4).

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5. DETAILED REQUIREMENTS

5.1 Composition. The specified lead composition (see 1.2.1.3) shall conform to the requirements of the applicable documents as follows:

A - Copper-iron-phosphorus-zinc alloy (194 alloy): ASTM B465.
B - Copper-iron-phosphorus-cobalt-tin alloy (195 alloy), 97 percent Cu, 1.5 percent Fe, 0.1 percent P, 0.8 percent Co, and 0.6 percent Sn.
C - Copper (oxygen free) ASTM B170, grade 2.
D - Copper-clad nickel-iron alloy: ASTM F29.
E - Copper-nickel-tin alloy (725 alloy) 88.2 percent Cu, 9.5 percent Ni, and 2.3 percent Sn: ASTM B122.
F - Iron-nickel alloy (52 alloy): ASTM F30.
G - Iron-nickel alloy (42 alloy): ASTM F30.
H - Copper-aluminum-silicon-cobalt alloy (638 alloy) 95.0 percent Cu, 2.8 percent Al, 1.8 percent Si, and 0.4 percent Co.
J - Copper-zinc-aluminum-cobalt alloy (688 alloy): ASTM B592.
K - Iron-nickel-cobalt alloy: ASTM F15.
L - Copper, Electrolytic tough - pitch: ASTM B5.
M - Steel, nickel clad: 8 percent nickel clad by weight.
N - Nickel: MIL-N-46025 (for ribbon leads) and MIL-N-46026 (for round-wire leads).
W - Steel, copper-clad: ASTM B452 (except that the percent of conductivity shall be extended to a minimum of 16 percent).
X - Other composition as specified in applicable acquisition document.

5.2 Final finish. The final finish for leads (see 1.2.1.4) shall conform to one of the following as applicable:

01 - No finish. Lead is supplied with no external finish.
21 - Gold-plating, electrodeposited. Gold plating shall be in accordance with MIL-G-45204, type I. The thickness of the gold-plating shall be as designated in 1.2.1.4. When specified, under-plating shall be in accordance with 5.2.1.1.
22 - Gold-plating, electrodeposited. Gold plating shall be in accordance with MIL-G-45204, type I. The thickness of the gold-plating shall be as designated in 1.2.1.4. When specified, under-plating shall be in accordance with 5.2.1.1.
31 - Acid-tin plating, electrodeposited. The acid-tin plating shall be in accordance with MIL-T-10727 and shall be a minimum of 300 microinches (7.62 µm) and a maximum of 500 microinches (12.7 µm) thick as per 1.2.1.4.
32 - Tin-lead plating, electrodeposited. The tin-lead plating shall be in accordance with MIL-P-81728 and shall have a composition of 50 to 70 percent tin (remainder lead). The thickness shall be a minimum of 300 microinches (7.62 µm) and a maximum of 900 microinches (22.9 µm).
33 - Matte-tin plating, electrodeposited. The matte-tin plating shall be in accordance with MIL-T-10727, type I. The thickness shall be a minimum of 300 microinches (7.62 µm) and a maximum of 500 microinches (12.7 µm) as per 1.2.1.4. The matte-tin plating shall contain no more than 0.05 percent by weight codeposited organic material measured as elemental carbon.
34 - Fused (reflowed) tin-lead plating, electrodeposited. The tin-lead plating shall be in accordance with MIL-P-81728 and shall have a composition of 50 to 70 percent tin (remainder lead). The tin-lead thickness after fusing (or reflow), shall not be less than 100 microinches (2.54 µm) at any single point. The average thickness of all points measured shall be a minimum of 300 microinches (7.62 µm) and a maximum of 500 microinches (12.7 µm).
35 - Acid-tin plating, electrodeposited. The acid-tin plating shall be in accordance with MIL-T-10727 and shall be a minimum of 100 microinches (2.54 µm) and a maximum of 300 microinches (7.62 µm) thick as per 1.2.1.4.

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36 - Matte-tin plating, electrodeposited. The matte-tin plating shall be in accordance with MIL-T-10727, type I. The thickness shall be a minimum of 100 microinches (2.54 µm) and a maximum of 300 microinches (7.62 µm) as per 1.2.1.4. The matte-tin plating shall contain no more than 0.05 percent by weight codeposited organic material measured as elemental carbon.

51 - Hot solder dip. The solder shall be composition Sn 60, Sn 63, or Sn 62 of QQ-S-571 and the coating shall have a minimum thickness as specified in 1.2.1.4. Hot solder dip finish may be used over final finish types 21, 22, 31, 32, 33, 34, 35, 36, 61, 71, and 72. When specified (see 1.2.1.5), electroless nickel underplating may be used in accordance with 5.2.1.3.

52 - Hot solder dip. The solder shall be composition Sn 60, Sn 63, or Sn 62 of QQ-S-571 and the coating shall have a minimum thickness as specified in 1.2.1.4. Hot solder dip finish may be used over final finish types 21, 22, 31, 32, 33, 34, 35, 36, 61, 71, and 72. When specified (see 1.2.1.5), electroless nickel underplating may be used in accordance with 5.2.1.3.

61 - Nickel plating, electrodeposited. The nickel plating shall be in accordance with QQ-N-290. The thickness shall be a minimum of 50 microinches (1.27 µm) and a maximum of 150 microinches (3.81 µm). The nickel plating shall be nonbrightened and low stress.

71 - Silver plating, electrodeposited. The silver plating shall be in accordance with QQ-S-365, type II, grade B. The thickness shall be a minimum of 150 microinches (3.81 µm) and a maximum of 350 microinches (8.89 µm).

72 - Silver plating, electrodeposited. The silver plating shall be matte, nonbrightened in accordance with QQ-S-365, type I. The thickness shall be a minimum of 150 microinches (3.81 µm) and a maximum of 350 microinches (8.89 µm).

5.2.1 Underplating requirements.

5.2.1.1 Gold plating. A copper plus electrodeposited nickel or electrodeposited nickel underplating is required on all leads with gold plating as the final finish in accordance with MIL-C-14550.

5.2.1.2 Compositions F, G, K. All leads with composition F, G, or K shall have an underplating prior to final finish.

5.2.1.3 Hot solder dip. When specified for use under hot solder dip lead finish (see 1.2.1.5), the electroless nickel underplating shall be in accordance with MIL-C-26074. The thickness shall be as designated in 1.2.1.5B.

5.2.1.4 Compositions A, B, J, X. All leads with compositions A, B, J, or X shall have a copper or nickel underplate of 100 microinches (2.54 µm) minimum.

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6. NOTES

6.1 Precaution. Substitution of any type of lead for the existing leads on electronic component parts should not be made in a parts specification without first determining the effect of this substitution on the performance and solderability characteristics of the component part.

6.2 Finishes. All plating of final finishes or underplating shall be deposited in such a way that the plating is applied to clean, nonoxidized metal surfaces that are free from defects that will be detrimental to the utility, form, fit, function, or protective value of the finish.

6.3 Removal of gold plating before soldering. Prior to equipment assembly, the gold plated leads shall have the gold removed before soldering by single or double dipping into a flowing or nonflowing hot solder of sufficient volume to assure complete gold removal.

6.4 Passive nickel. If there is a delay after the application of nickel strike, where the nickel surface may have become passive, the nickel shall be reactivated followed by gold plating. Reactivation of the nickel surface should be in accordance with the appropriate section of the latest ASTM standards. (Reference: ASTM B322, Cleaning Metals Prior to Electroplating; and ASTM B343, Preparation of Nickel for Electroplating with Nickel.)

6.5 Cross-reference. The cross-reference for the discontinued and superseding lead types is in accordance with table III. Where MIL-STD-1276B designations are specified, the cross referenced MIL-STD-1276C designations may be used.

TABLE III. Cross-reference of lead types.

Superseding type			Discontinued type
MIL-STD-1276C			MIL-STD-1276B
Composition	Final finish	Underplating	
C	3	---	C (tin plated)
C	4	---	C (tin-lead plated)
D	2a	A1	D
F	2a	A1	F
K	2a	A1	K
N	1	---	N-1
N	2a	---	N-2
N	3	---	N-3 (tin plated)
N	4	---	N-3 (tin-lead plated)

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TABLE III. Cross-reference of lead types - Continued.

MIL-STD-1276D	MIL-STD-1276C
Composition	Composition
A	Not available
B	Not available
C	C
D	D
E	Not available
F	F
G	G
H	Not available
J	Not available
K	K
L	C_1
M	Not available
N	N
W	W
X	X
Underplating (same as revision C with the addition of C and P)	
Final finish	Final finish
01	1
21	2a
22	2b
31	3
32	4
33	Not available
34	Not available
51	5a
52	5b
61	6
71	7
72	Not available

6.6 Tin melting. Flow-melting or reflowing of tin deposits should be limited to electrodeposited thicknesses of 100 microinches (2.54 µm) minimum and 300 microinches (7.62 µm) maximum. The melting of tin deposits in excess of 300 microinches (7.62 µm) tend to ball-up or dewet.

6.7 Cleanliness. The last step in any component manufacturing should be cleaning to remove residues such as fluxes.

6.8 Subject term (key word) listing.

Final finish
Leads
Plating
Underplating

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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